Adjudicating the cylindrical negative-energy blowup claim for biharmonic NLS

Mathematical verification note

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Abstract

We identify the failure of the final virial inference in Theorem 1.2 of Gou's publicly accessible arXiv version 3. An elementary family of smooth negative-energy cylindrical test functions, including a zero-momentum family, disproves the uniform virial control required by the cited argument. This observation alone is a proof obstruction. The stronger refutation of the theorem's statement in dimension four follows from our separately validated negative-energy traveling profiles. We rederive the signs, prove the finite axial moment, and record a fresh execution of all nine foundational verification programs. The counterexamples apply both to zero and to every prescribed positive coefficient of the second-order dispersion.

Version and status. The statement checked here is [1, Theorem 1.2], arXiv:2205.08167v3, dated 4 November 2024. The publisher confirms the corresponding journal publication, but its full final text was not accessible in this check; we do not claim a line-by-line comparison of that version. This note records a mathematical adjudication using the supplied computer-assisted proof, not an official correction issued by the author or journal. The fresh runs reproduce the supplied verifiers in a separate directory; they are not an independent reimplementation or an external review.

1 The precise incompatible statements

Write $x = (y, s) \in \mathbb{R}^3 \times \mathbb{R}$. We use

$$\mathrm{i}u_t = \Delta^2 u - \mu\Delta u - |u|^2 u, \qquad \mu \geq 0, \tag{1}$$

$$E_\mu(u) = \frac{1}{2}\|\Delta u\|_2^2 + \frac{\mu}{2}\|\nabla u\|_2^2 - \frac{1}{4}\|u\|_4^4, \quad M(u) = \|u\|_2^2, \quad P_s(u) = \operatorname{Im}\int \bar{u}\partial_s u.$$

Gou's Theorem 1.2 includes dimension four and assumes precisely cylindrical symmetry, $u_0 \in H^2$, $su_0 \in L^2$, and $E_\mu(u_0) < 0$. It asserts finite-time blowup for $\mu > 0$, and finite- or infinite-time blowup for $\mu = 0$. No zero-momentum, ground-state, or uniform-in-time axial-moment hypothesis is imposed.

The verified construction used below supplies smooth cylindrical traveling profiles U with $sU \in L^2$ and $E_\mu(U) < 0$. The exact solution

$$u(t, y, s) = e^{\mathrm{i}\omega t} U(y, s - vt) \tag{2}$$

exists for every real time and has constant finite H^2 norm. Thus it contradicts the $\mu = 0$ conclusion in its standard energy-space grow-up meaning. The $\mu > 0$ examples contradict a finite-time conclusion directly, independently of any possible ambiguity in the phrase "infinite-time blowup." No multiwave gluing result is needed for this contradiction.

2 The precise failure of the cited virial inference

Gou's multiplier and virial quantity are

$$\phi_R(y,s) = \psi_R(|y|) + \frac{s^2}{2}, \qquad \mathcal{M}_R(u) = 2\,\mathrm{Im}\int \bar{u}\,\nabla\phi_R\cdot\nabla u. \tag{3}$$

The final sentence after equations (2.25) and (2.29) invokes the proof of Boulenger–Lenzmann's Theorem 3. For a fully localized multiplier, the crucial step in that proof, equation (7.15), uses the finite quantity $\|\nabla\phi_R\|_\infty$ to obtain a bound of the form

$$|\mathcal{M}_R(u)| \le 2\,\|\nabla\phi_R\|_\infty\,\|u\|_2^{3/2}\,\|\Delta u\|_2^{1/2}. \tag{4}$$

Together with a uniform negative upper bound on $\mathcal{M}_R'$, this forces $\|\Delta u(t)\|_2$ to grow. For (3), however, $\partial_s\phi_R = s$, so the coefficient in (4) is infinite. The applicable estimate is instead

$$|\mathcal{M}_R(u)| \le 2\,\|\nabla_y\psi_R\|_\infty\,\|u\|_2\,\|\nabla_y u\|_2 + 2\,\|su\|_2\,\|\partial_s u\|_2\,. \tag{5}$$

Initial finiteness of $\|su\|_2$ is neither its conservation nor a uniform-in-time bound. Consequently the last inference cannot be transferred. This is an explicit failure of a necessary hypothesis of the cited step; it is not a criticism based merely on an abbreviated proof.

Proposition 1 (Elementary disproof of uniform virial control)**.** *There are smooth compactly supported cylindrical functions w_a such that $P(w_a) = 0$, $E_\mu(w_a) < 0$, and the mass, energy and H^2 norm are independent of a, while $\mathcal{M}_R(w_a) \to -\infty$ as $a \to +\infty$.*

Proof. Choose a nonzero real $f \in C_c^\infty(\mathbb{R}^4)$, radial in y and even in s, supported in $|s| < b$. Fix $k < 0$ and choose $A > 0$ sufficiently large that

$$e := \frac{A^2}{2}\left(\left\|\Delta(e^{\mathrm{i}ks}f)\right\|_2^2 + \mu\left\|\nabla(e^{\mathrm{i}ks}f)\right\|_2^2\right) - \frac{A^4}{4}\,\|f\|_4^4 < 0.$$

For $a > b$ set

$$w_a(y,s) = A\left[e^{\mathrm{i}ks}f(y,s-a) + e^{-\mathrm{i}ks}f(y,s+a)\right], \qquad m = \|f\|_2^2\,.$$

The two supports are disjoint, including the supports of all derivatives. Translation invariance and conjugation give

$$M(w_a) = 2A^2 m, \qquad E_\mu(w_a) = 2e < 0, \qquad P(w_a) = 0,$$

and the standard H^2 norm is independent of a. The transverse current vanishes. On the two packets the axial currents are $kA^2 f(y,s-a)^2$ and $-kA^2 f(y,s+a)^2$, respectively. Evenness gives

$$\mathcal{M}_R(w_a) = 4kaA^2 m \longrightarrow -\infty. \tag{6}$$

Every w_a has finite variance. The functions are also inversion-even. This proves the assertion. □

The parameter a in Proposition 1 is a spatial separation, not an asserted solution time. These test functions disprove the functional estimate; by themselves they do not disprove the dynamical theorem. The next sections supply the exact dynamical counterexample.

3 Fresh verification of the four-dimensional profile input

Set $A_0 = \Delta^2 + 1$ and $\|h\|_Z^2 = \|\Delta h\|_2^2 + \|h\|_2^2$. The foundational manuscript [3] defines a compact rational Hermite profile q and a compact rational response ψ. The fresh check uses the unchanged profile with SHA-256

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c50a6f98682ce5f5bef9ab62ae2a93ba06b9
1522c8f3e29b859ac521afb8f627.
```

The following programs were executed from the frozen inputs in a separate directory: one exact profile/Ritz verifier; one exact response verifier; one high-angular verifier; one rational implication verifier; and five interval Birman–Schwinger verifiers. All nine terminated successfully. The comparison algorithms and their analytic interfaces were also inspected, rather than treating program return codes as proofs.

Channel (c, ℓ)	Quantity	Fresh strict upper bound
$(3, 0)$	$\|K_{c,q,\ell}\|_{\mathrm{HS}}^2$	$38.795691 < 38.80$
$(1, 1)$	$\operatorname{tr} K_{c,q,\ell}$	$1.691185 < 1.70$
$(3, 2)$	$\|K_{c,q,\ell}\|_{\mathrm{HS}}^2$	$4.797953 < 4.80$
$(1, 3)$	$\|K_{c,q,\ell}\|_{\mathrm{HS}}^2$	$0.211703 < 0.22$
$(3, 4)$	$\|K_{c,q,\ell}\|_{\mathrm{HS}}^2$	$0.725270 < 0.73$

Here $K_{c,q,\ell}$ is the restriction of $A_0^{-1/2} c q^2 A_0^{-1/2}$ to angular degree ℓ. The displayed decimals are conservative summaries; actual comparisons use rational or outward-rounded ball arithmetic. Exact Ritz lower bounds are $5.395, 2.999$ in the $(3, 0)$ channel, 2.11 in $(3, 2)$, and 1.04 in $(1, 1)$. The trace/HS budgets bound the entire omitted radial spectrum. Direct angular form monotonicity bounds all higher required angular degrees.

The resulting radial inverse and nonlinear contraction give an exact real stationary state Q, with

$$A_0 Q - Q^3 = 0, \qquad \|Q - q\|_Z \le \rho = \frac{3}{500}.$$

The original, less sharp radius is already sufficient. On the required parity space, $L_+ = A_0 - 3Q^2$ on real even degrees and $L_- = A_0 - Q^2$ on imaginary odd degrees are both invertible. In degree one,

$$\left\| L_-^{-1} \right\|_{Z^* \to Z} \le \gamma^{-1}, \qquad \gamma = \frac{2247991}{64000000} > 0.$$

For $g = \partial_s Q$, the response identity

$$\left\langle g, L_-^{-1} g \right\rangle = 2 \langle g, \psi \rangle - \langle \psi, L_- \psi \rangle + \left\langle g - L_- \psi, L_-^{-1} (g - L_- \psi) \right\rangle$$

is valid even though L_- is indefinite. Including the profile, potential and source transfers gives the exact rational bound

$$I := \left\langle \partial_s Q, L_-^{-1} \partial_s Q \right\rangle \le -\frac{24609689664617}{140499437500} < -175 < 0. \tag{7}$$

Thus the key sign comes from a full inverse-pairing certificate, not from an approximate eigenvalue or merely a negative trial quadratic form. The computational trusted base is the documented exact rational arithmetic and Python-FLINT/Arb outward enclosures. Source code, frozen data and fresh outputs accompany this note.

4 Signs, exact solutions and the finite axial moment

For $\mu = 0$, direct substitution of (2), with $\omega = 1$, gives

$$A_0 U + \mathrm{i}v\partial_s U - |U|^2 U = 0. \tag{8}$$

The real gradient of P_s is $-2\mathrm{i}\partial_s U$; hence the action is $E_0 + M/2 - vP_s/2$. On cylindrical functions satisfying $U(y,-s) = \overline{U(y,s)}$, its profile derivative at $(v,U) = (0,Q)$ is $L_+ \oplus L_-$. The parity gaps above remove both symmetry kernels and any accidental kernels. The analytic implicit-function theorem supplies

$$U_v = Q - \mathrm{i}vL_-^{-1}\partial_s Q + O_Z(v^2), \qquad P_s(U_v) = 2Iv + O(v^3).$$

The mass-preserving dilation $U_\lambda(x) = \lambda^2 U(\lambda x)$ gives the Pohozaev identity

$$4E_0(U_v) - \frac{v}{2}P_s(U_v) = 0, \qquad E_0(U_v) = \frac{I}{4}v^2 + O(v^4) < 0 \tag{9}$$

for sufficiently small nonzero v. The dilation identity can be justified first with a compact cutoff; its errors are unweighted H^2 tails. These signs have been independently rederived.

Lemma 1 (Finite moments required by the cylindrical theorem)**.** *For sufficiently small $|v|$, the profiles just constructed are smooth and satisfy $x_j U_v \in H^4$ for every coordinate j.*

Proof. Since $H^2(\mathbb{R}^4) \subset L^6$, (8) first gives $U \in H^4$. Elliptic bootstrap then gives all unweighted Sobolev orders, and $U(x) \to 0$ at infinity. The free operator $D = \Delta^2 + 1 + \mathrm{i}v\partial_s$ has symbol $|\xi|^4 + 1 - v\xi_s$, bounded below by $1 - 3(|v|/4)^{4/3} > 0$. Therefore $\|w\|_{H^4} \le C_v \|Dw\|_2$.

Let $f_R = x_j\chi(x/R)$, where $\chi = 1$ near zero is smooth and compactly supported. Its positive-order derivatives through order four are uniformly bounded for $R \ge 1$. Consequently $[D, f_R]U$ is uniformly bounded in L^2 by the already available H^3 norm. Also

$$D(f_R U) = |U|^2 f_R U + [D, f_R]U.$$

Choose L such that $\sup_{|x|>L}|U|^2 < (2C_v)^{-1}$. In $|x| \le L$, $|f_R| \le CL$ uniformly in R; outside, the potential term can be absorbed into the left side of the elliptic bound. This gives a uniform H^4 bound on $f_R U$. Weak compactness and its local limit yield $x_j U \in H^4$. The same argument applies to $D = \Delta^2 - \mu\Delta + 1 + \mathrm{i}v\partial_s$ for small parameters with a positive free symbol. □

Thus $U_v \in \Sigma_4$ exactly as required. Substitution gives a global solution with constant H^2 norm; no local-to-global continuation conjecture is involved.

For this exact solution, change variables $s \mapsto s - vt$ in (3) to obtain

$$\mathcal{M}_R(u(t)) = \mathcal{M}_R(U_v) + 2vtP_s(U_v), \qquad \mathcal{M}_R'(u(t)) = 16E_0(U_v) < 0. \tag{10}$$

The density of U_v is even in s, so

$$\|su(t)\|_2^2 = \|sU_v\|_2^2 + v^2t^2M(U_v). \tag{11}$$

The negative virial slope is therefore perfectly consistent with bounded H^2 evolution. The growing quantity is the axial spatial moment. Equations (10)–(11) identify the failure dynamically, in addition to the functional obstruction of Proposition 1.

5 The positive second-order coefficient is also covered

The two-parameter profile map

$$(\mu, v, U) \longmapsto (\Delta^2 - \mu\Delta + 1)U + \mathrm{i}v\partial_s U - |U|^2 U$$

is analytic from the same parity space to its dual; its profile derivative at $(0, 0, Q)$ is the already verified invertible operator. The resulting branch $U_{\mu,v}$ satisfies the mixed-dispersion Pohozaev identity

$$E_\mu(U_{\mu,v}) = \frac{v}{8} P_s(U_{\mu,v}) + \frac{\mu}{4} \|\nabla U_{\mu,v}\|_2^2 . \tag{12}$$

Choose $0 < \eta < -I/\|\nabla Q\|_2^2$ and put $\mu = \eta v^2$. Continuity of the branch gives

$$E_{\eta v^2}(U_{\eta v^2, v}) = \frac{1}{4}\left(I + \eta \|\nabla Q\|_2^2\right) v^2 + o(v^2) < 0.$$

This supplies at least one example with $\mu_0 > 0$. For any prescribed $\mu_* > 0$, set $\lambda = \sqrt{\mu_*/\mu_0}$ and $u_*(t, x) = \lambda^2 u_0(\lambda^4 t, \lambda x)$. Then u_* solves (1) with μ_* and

$$E_{\mu_*}(u_*(0)) = \lambda^4 E_{\mu_0}(u_0(0)) < 0.$$

Its frequency is λ^4, its velocity is $\lambda^3 v_0$, its H^2 norm is constant in time, and its axial moment is finite at each finite time. The frequency is free in Gou's Cauchy problem, so this scaling does not violate any stated hypothesis.

6 Disposition and scope

The final virial inference in the checked public version is invalid: a uniform negative derivative of an axially unbounded virial does not force growth of the unweighted energy norm. This fact is proved without any computer-assisted premise by the packet calculation.

The separate exact-profile construction, whose foundational finite computations were rerun and whose analytic interfaces were reviewed here, supplies the stronger conclusion: in dimension four, Theorem 1.2(ii) of that version is false under the standard H^2 grow-up interpretation, and Theorem 1.2(i) is false for every prescribed $\mu > 0$ when the profile frequency is unrestricted. A counterexample at one allowed dimension suffices to refute each universal statement. No assertion about every other fixed dimension or about Theorem 1.1 is made.

These examples are nonradial and do not contradict the radial theorems of Boulenger–Lenzmann or the later radial finite-time result. They do not establish orbital stability. Minor coefficient or indexing errors elsewhere in Gou's proof are unnecessary to this adjudication and are not used as substitutes for the counterexample.

References

[1] T. Gou, *Blowup of cylindrically symmetric solutions for biharmonic NLS*, arXiv:2205.08167v3, 4 November 2024. Journal metadata: Proc. Edinburgh Math. Soc. 67 (2024), 1085–1098, doi:10.1017/S0013091524000664. The inspected full text is the arXiv version.

[2] T. Boulenger and E. Lenzmann, *Blowup for biharmonic NLS*, Ann. Sci. Ec. Norm. Super. 50 (2017), 503–544; arXiv:1503.01741, especially Theorem 3 and equation (7.15).

[3] *Global negative-energy traveling waves for the mass-critical biharmonic Schrodinger equation in four space dimensions*, internal computer-assisted research manuscript, 7 October 2026. File `BNLS_Nonradial_Research_20261007.tex`; supplied with the exact frozen data and verification programs. This note rechecks its foundational counterexample and adds an explicit adjudication of the cylindrical claim.